

Notation

The big issue in choosing notation for a book on supersymmetry is whether to use a two-component or a four-component notation for spinors. The standard texts on supersymmetry have opted for the two-component Weyl notation. I have chosen instead to use the four-component Dirac notation except in the first stages of constructing the supersymmetry algebra and multiplets, because I think this will make the book more accessible to those physicists who work on particle phenomenology and model building. It would be a pity to see the growth of a separate enclave of supersymmetry specialists, who communicate well with each other but are cut off by their notation from the larger community of particle theorists.

There is no great difficulty anyway in converting expressions in four-component form into the two-component formalism. In the representation of the Dirac matrices used throughout this book, in which γ_5 is the diagonal matrix with elements $+1, +1, -1$, and -1 on the main diagonal, any four-component Majorana spinor ψ_α (such as the supersymmetry generator Q_α , the superspace coordinate θ_α , or the superderivative $\mathcal{D}_\alpha$) may be written in terms of a two-component spinor χ_a as

$$\psi = \begin{pmatrix} e\chi^* \\ \chi \end{pmatrix},$$

where e is the $2 \cdot 2$ antisymmetric matrix with $e_{12} = +1$. The two-component spinor χ_a is what in other books is often called $\psi_a^* = \bar{\psi}_{\dot{a}}$, while $(e\chi^*)_a$ would be called ψ^a . A summary of useful properties of four-component Majorana spinors is given in the appendix to Chapter 26.

Here are some other features of the notation used in this book:

Latin indices i, j, k , and so on generally run over the three spatial coordinate labels, usually taken as $1, 2, 3$. Where specifically indicated, they run over values $1, 2, 3, 4$, with $x^4 = it$.

Greek indices μ, ν , etc. from the middle of the Greek alphabet generally run over the four spacetime coordinate labels $0, 1, 2, 3$, with x^0 the time coordinate. Where it is necessary to distinguish between spacetime coordinates in a general coordinate system and in a locally inertial system, indices μ, ν , etc. are used for the former and a, b , etc. for the latter.

Greek indices α, β , etc. from the beginning of the Greek alphabet generally (except in Chapter 24) run over the components of four-component spinors. To avoid confusion, I depart here from the conventions of Volume II, and use upper-case letters A, B , etc. to label the generators of a symmetry algebra. Components of two-component spinors are labelled with indices a, b , etc. In particular, four-component supersymmetry generators are denoted Q_α , while two-component generators (the bottom two components of Q_α) are called Q_a .

Repeated indices are generally summed, unless otherwise indicated.

The spacetime metric $\eta_{\mu\nu}$ is diagonal, with $\eta_{\mu\nu} = \text{diag}(-1, +1, +1, +1)$.

The d'Alembertian is defined as $\square \equiv \eta^{\mu\nu} \partial^2 / \partial x^\mu \partial x^\nu = \nabla^2 - \partial^2 / \partial t^2$, where ∇^2 is the Laplacian $\partial^2 / \partial x^i \partial x^i$.

The ‘Levi–Civita tensor’ $\epsilon^{\mu\nu\rho\sigma}$ is defined as the totally antisymmetric quantity with $\epsilon^{0123} = +1$.

Dirac matrices γ^μ are defined so that $\gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu = 2\eta^{\mu\nu}$. Also, $\gamma_5 = i\gamma^0\gamma^1\gamma^2\gamma^3$, and $\beta = i\gamma^0 = \gamma_4$. Where explicit matrices are needed, they are given by the block matrices

$$\gamma^0 = -i \begin{pmatrix} 0 & \mathbf{1} \\ \mathbf{1} & 0 \end{pmatrix}, \quad \gamma^i = -i \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix},$$

where $\mathbf{1}$ is the unit $2 \cdot 2$ matrix, 0 is the $2 \cdot 2$ matrix with elements zero, and the components of σ are the usual Pauli matrices

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

We also frequently make use of the $4 \cdot 4$ block matrices

$$\gamma_5 = \begin{pmatrix} \mathbf{1} & 0 \\ 0 & -\mathbf{1} \end{pmatrix}, \quad \epsilon = \begin{pmatrix} e & 0 \\ 0 & e \end{pmatrix},$$

where e is again the antisymmetric $2 \cdot 2$ matrix $i\sigma_2$. For instance, our phase convention for four-component Majorana spinors s may be expressed as $s^* = -\beta\gamma_5\epsilon s$.

The step function $\theta(s)$ has the value $+1$ for $s > 0$ and 0 for $s < 0$.

The complex conjugate, transpose, and Hermitian adjoint of a matrix or vector A are denoted A^* , $A^\top$, and $A^\dagger = (A^*)^\top$, respectively. We use a dagger for the Hermitian adjoint of an operator and an asterisk for the complex conjugate of a number. For a vector or matrix of operators, the dagger also denotes the transpose of the matrix formed from the Hermitian adjoints of its components. $+H.c.$ or $+c.c.$ at the end of an expression indicates the addition of the Hermitian adjoint or complex conjugate of the foregoing terms. A bar on a four-component spinor u is defined by $\bar{u} = u^\dagger\beta$.

Units are used with $\hbar$ and the speed of light taken to be unity. Throughout, $-e$ is the rationalized charge of the electron, so that the fine structure constant is $\alpha = e^2/4\pi \simeq 1/137$. Temperatures are in energy units, with the Boltzmann constant taken equal to unity.

Numbers in parenthesis at the end of quoted numerical data give the uncertainty in the last digits of the quoted figure. Where not otherwise indicated, experimental data are taken from ‘Review of Particle Physics,’ The Particle Data Group, *European Physics Journal C* **3**, 1 (1998).